

Exploration of MRM Thermal Dynamics

Chris Schwarze, Ankit Maisuriya, Aaron Buikema
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Motivation

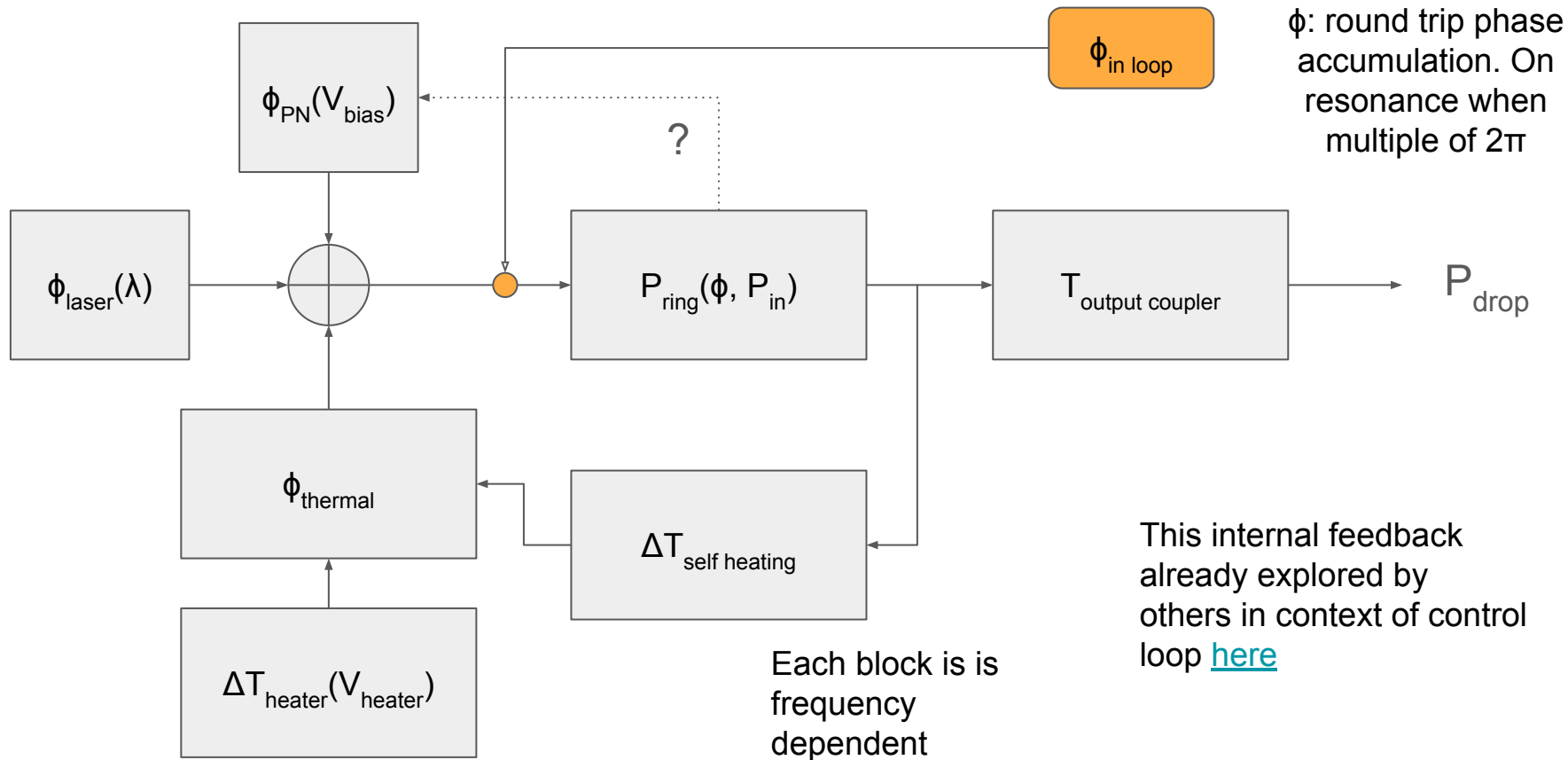
Concern that there may be high-frequency thermal effects that will affect bit stream, at much higher frequencies than previously assumed from our systems

Self Heating

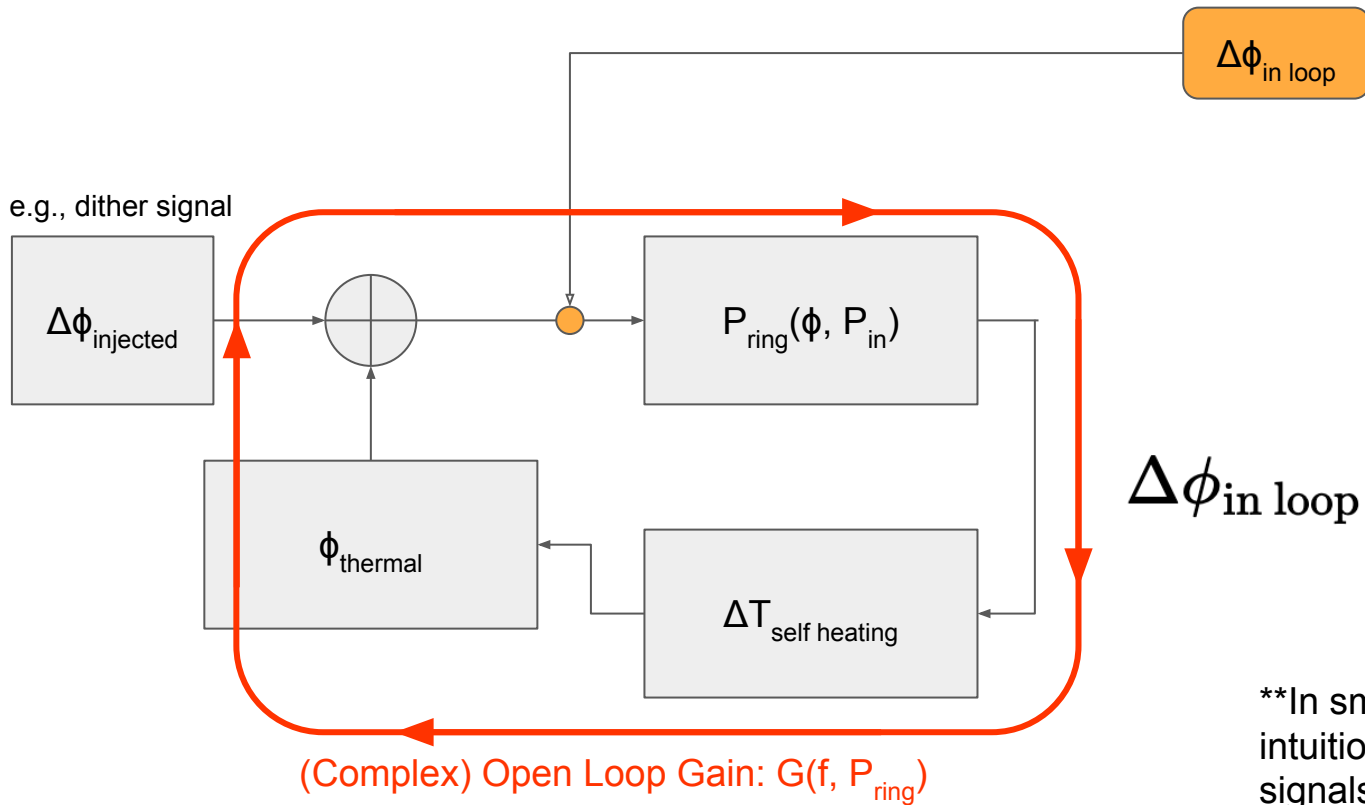
Idea: as get closer to resonance, more optical power $\Rightarrow$ more light absorbed in ring $\Rightarrow$ more heat applied, which shifts resonance of MRM

Depending on which side of resonance, will either be a negative- or positive-feedback mechanism. Operate on blue side, where feedback is negative. **This means that in general, intended drive will be suppressed.**

Inherent Feedback Loop



Inherent Feedback Loop



$$\Delta\phi_{\text{in loop}} = \Delta\phi_{\text{inj}} \frac{1}{1 - G}$$

**In small-signal regime,
intuition works for large
signals

Modeling

Ongoing modeling with PxV [here](#)
 Virtuoso sims [here](#)

Aaron wrote up simple modeling doc [here](#); let's review some important conclusions

Idea: MRM has “bare” Lorentzian response $\gamma_0(\lambda)$ with resonance wavelength λ_0 . We attempt to move resonance wavelength by λ_{inj} . Laser power P_L and ring power P_{ring} . G is self-heating loop gain (frequency dependent!)

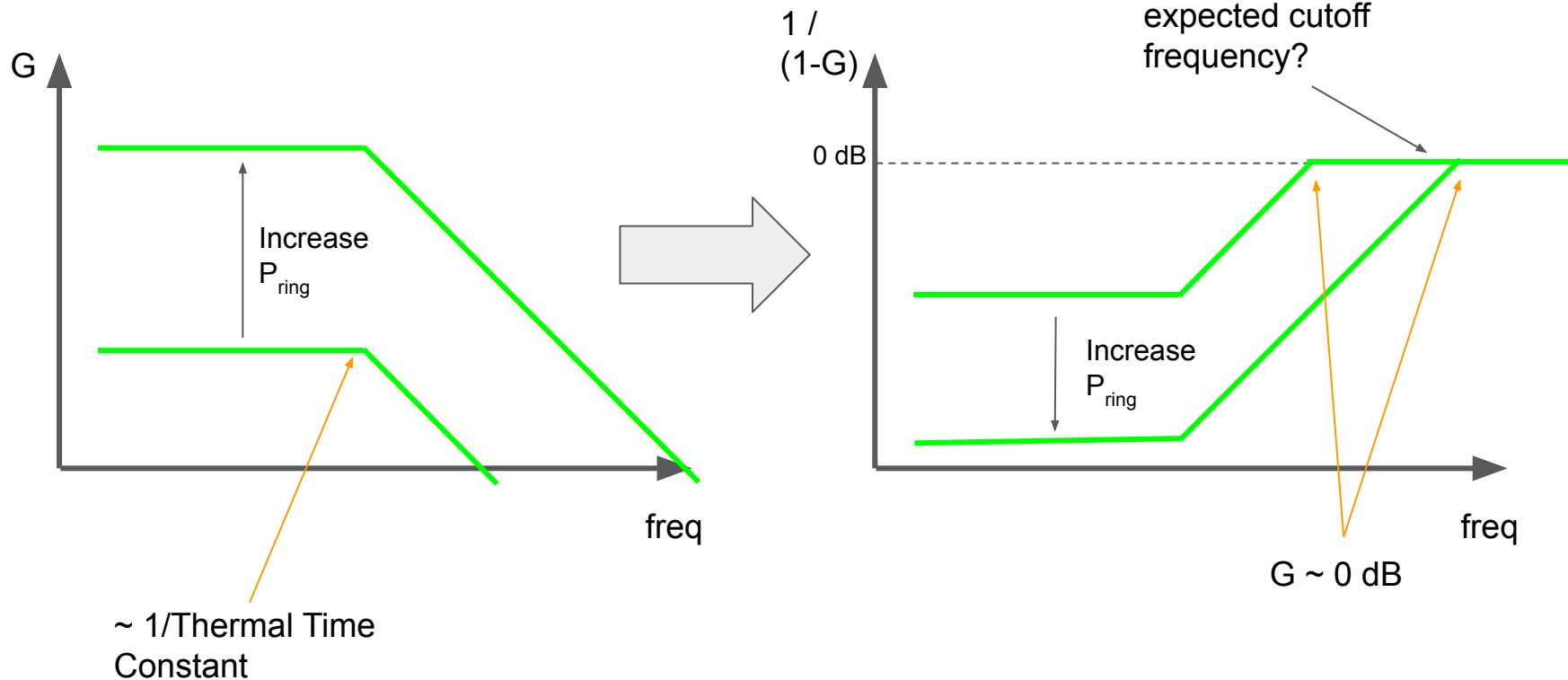
$$P_{ring} = P_L \gamma_0(\lambda)$$

$$\gamma_0(\lambda) = \frac{A}{1 + 4 \left(\frac{\lambda - \lambda_0}{\Delta\lambda} \right)^2}$$

$$\delta\lambda_0 = \delta\lambda_{inj} \frac{1}{1 - G}$$

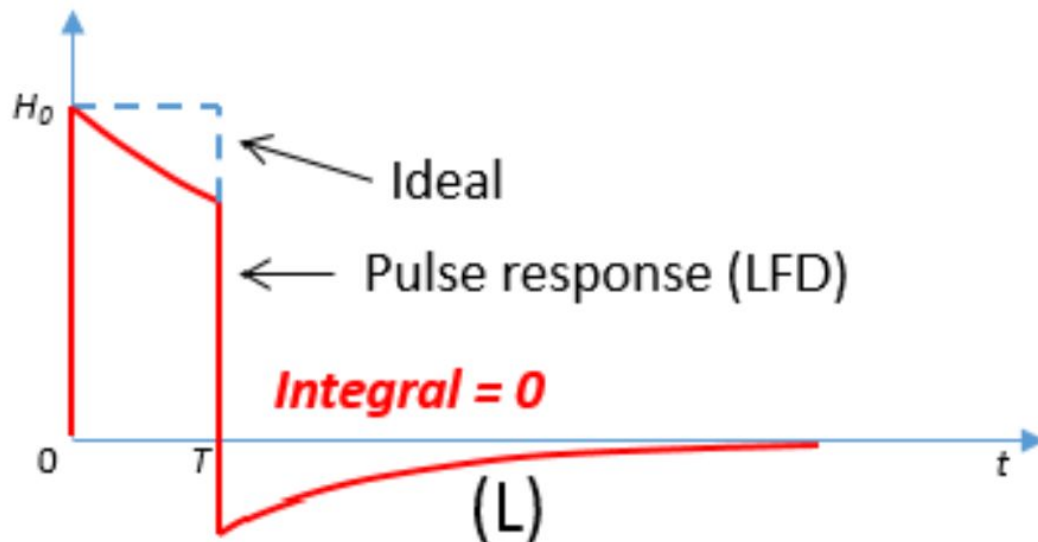
$$G(f) = B(f) P_L \frac{\partial \gamma_0}{\partial \lambda_0}$$

Inherent Feedback Loop



Effect on Link – Time Domain

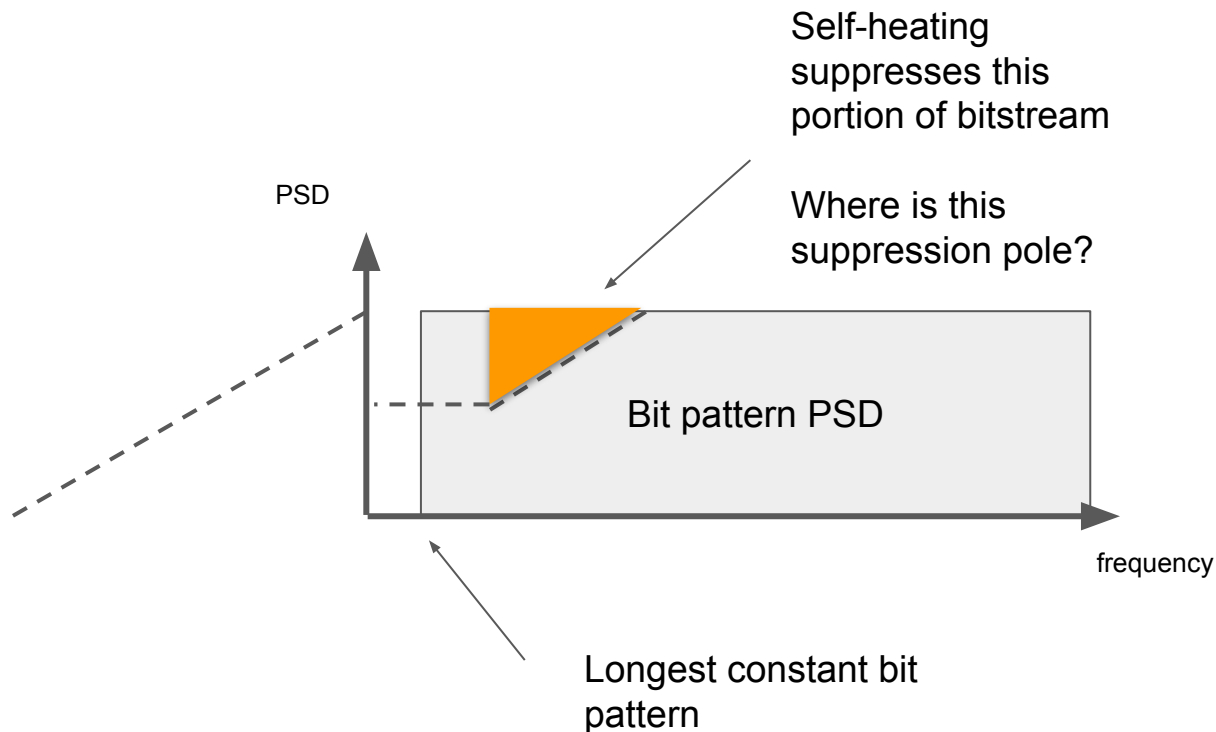
Causes baseline wander at low frequencies
Worse for longer bit patterns



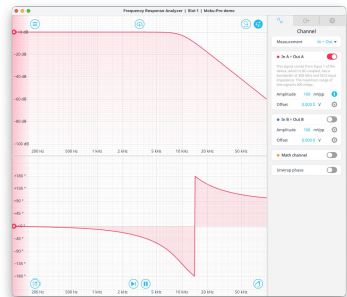
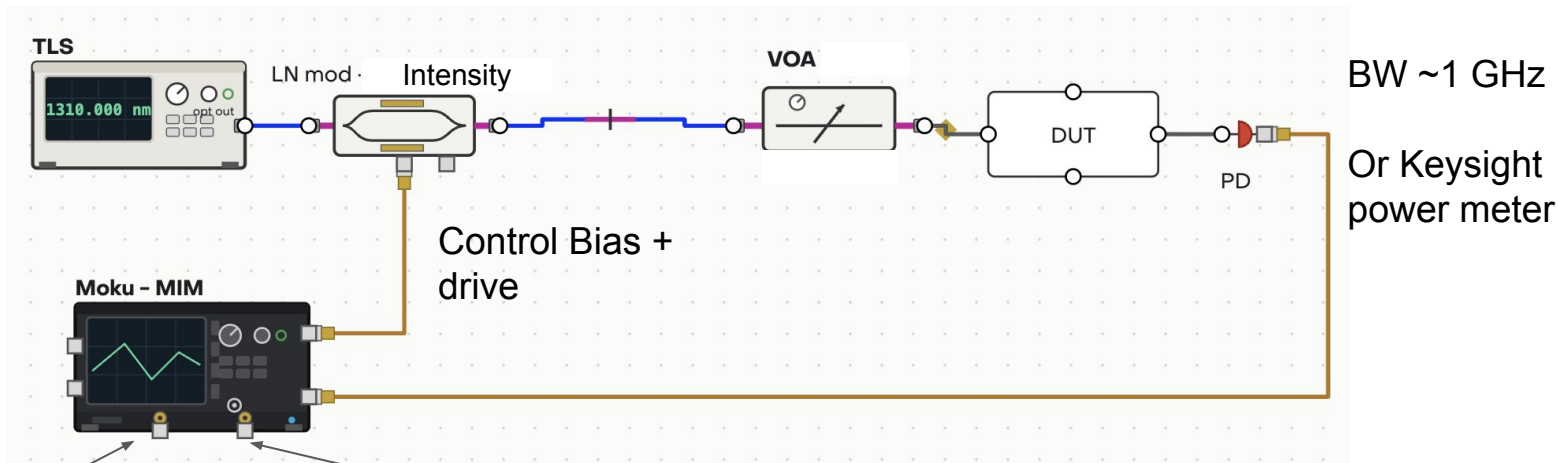
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Effect on Link – Frequency Domain

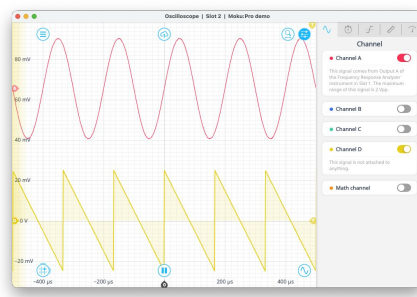
Similar to a high-pass effect



Experimental Setup



Frequency Response



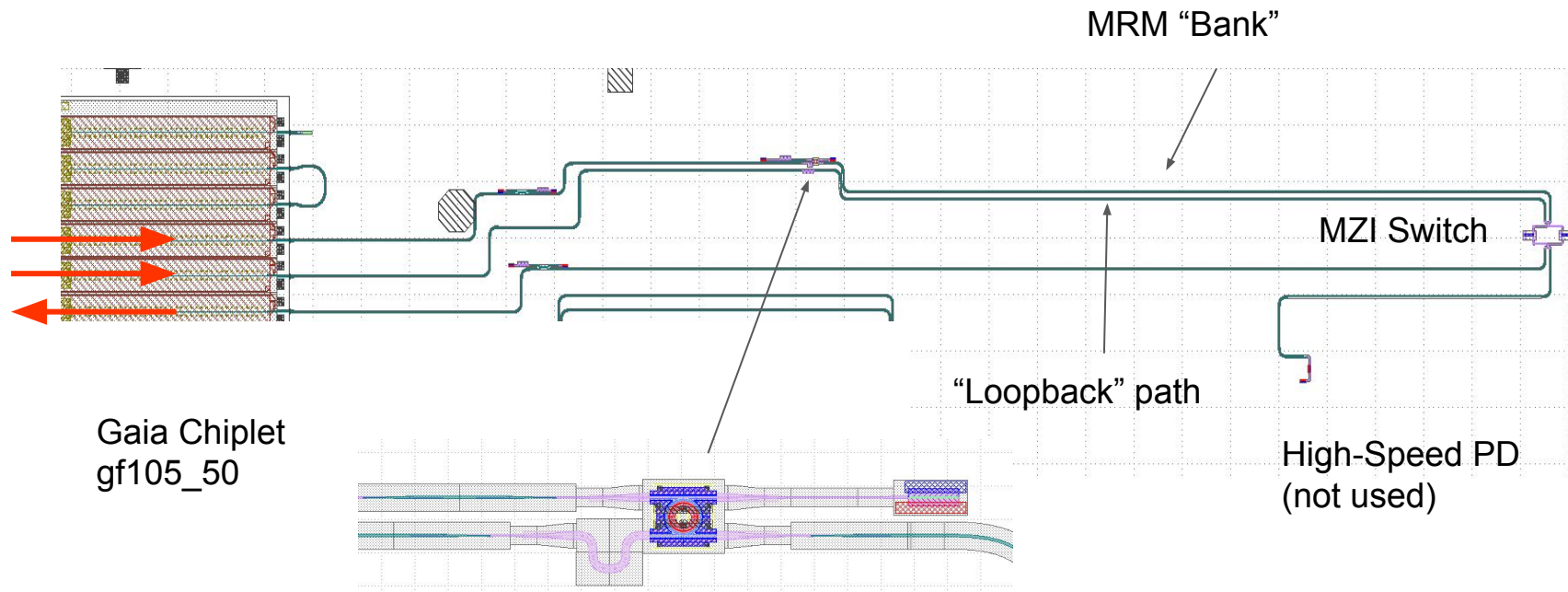
Time Series/Scope

DUT can be plain fiber, chip loopback, or MRM bank (on or off-resonance)

Measure response of PD vs. modulation drive vs. frequency up to 500 MHz (Moku: Pro limit) → small signal regime

Idea: See how MRM affects “simple” intensity modulation response

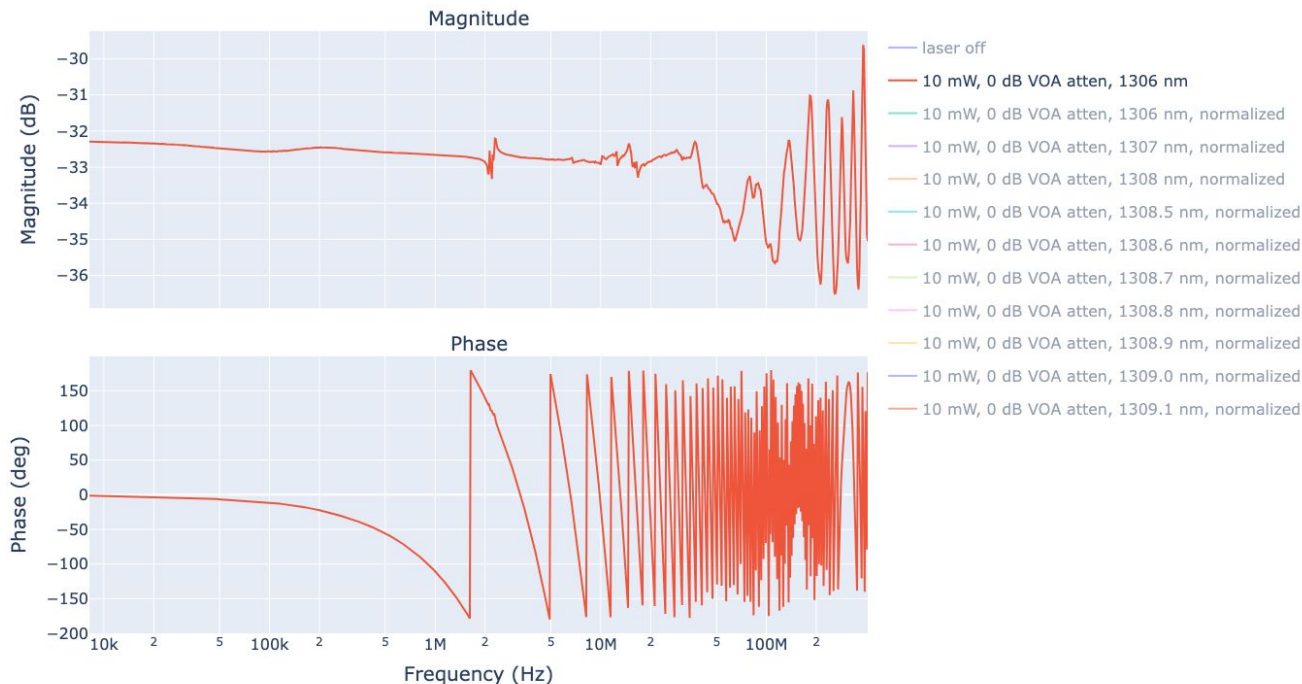
Experimental Setup - Gaia gf105_50



Results – Sanity Checks

Loopback Path (Raw)

Bode sweep overlay

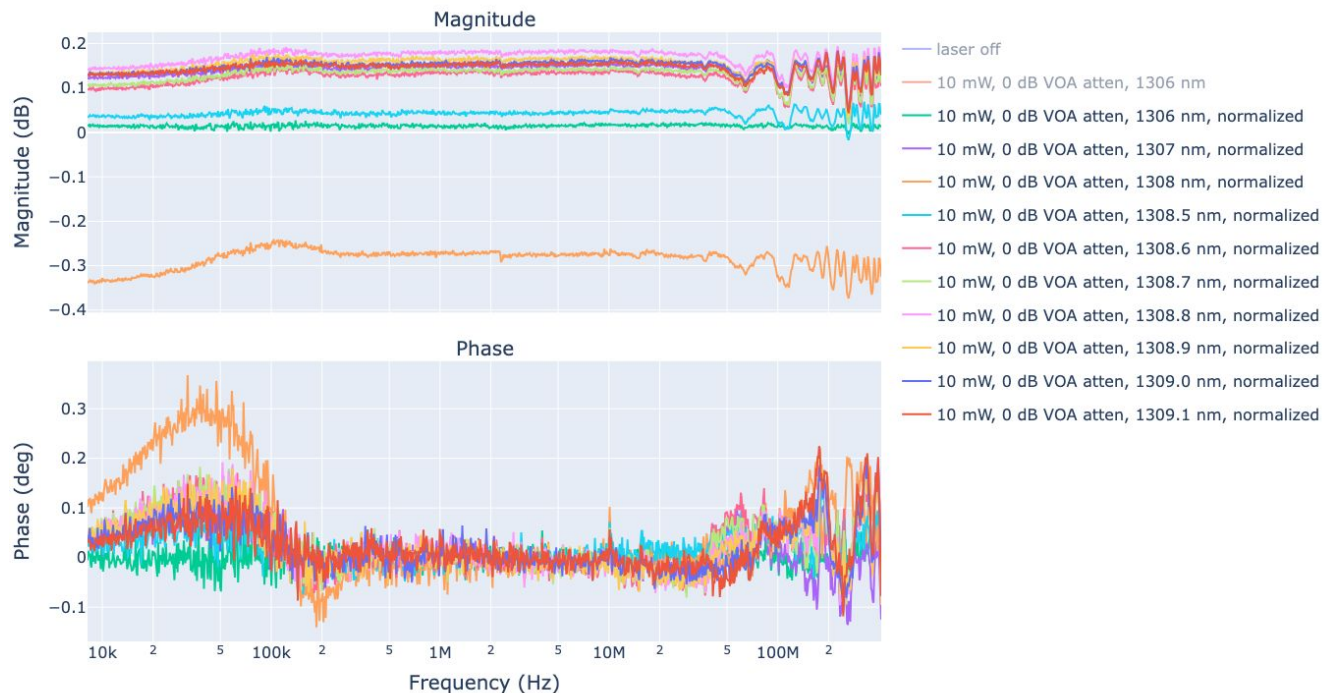


Observe system time delay and some ripple, but as long as constant, can de-embed

Results – Sanity Checks

Loopback Path (Normalized to raw response on previous slide)

Bode sweep overlay

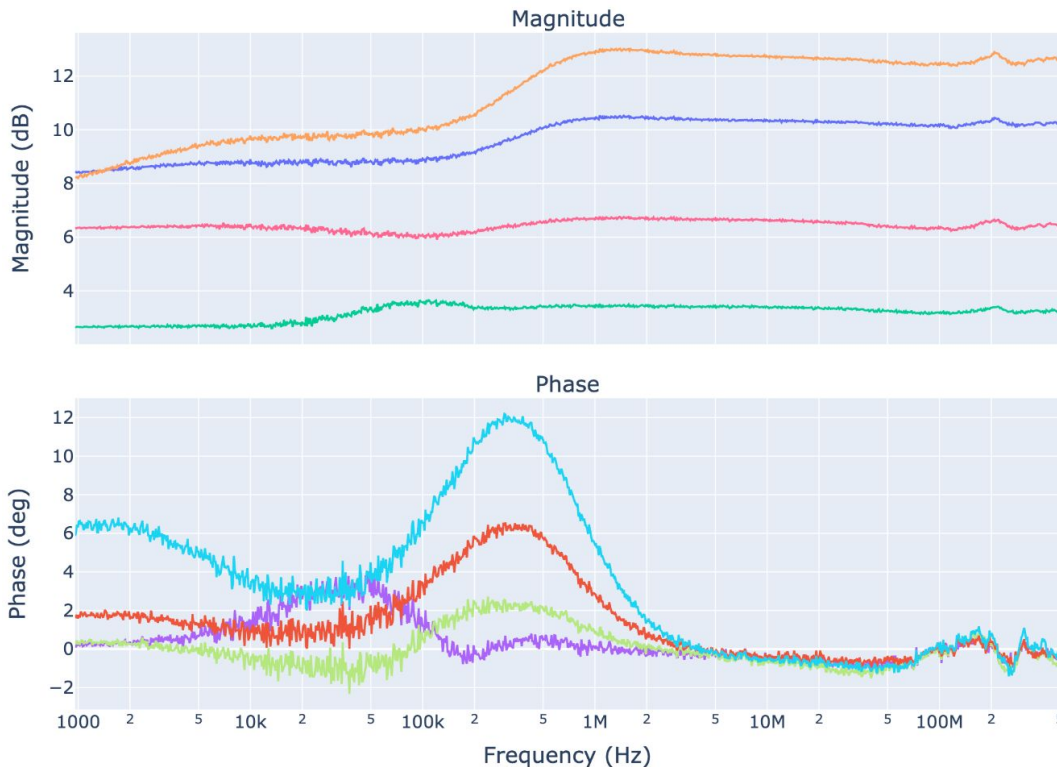


Conclusions:

- **Measurements are consistent**
- **Some variation in overall magnitude, likely from MZI or MZM or edge coupler wavelength variation**
- **(Not shown): PD is linear and flat across power levels**

Results – Self-heating observed

Bode sweep overlay



MRM PN junction NOT biased

Note: as power increases and wavelength goes up, detuning goes farther off resonance

Current traces: colors for phase and magnitude don't match....

- 15 mW in, no atten, 1308 nm, Set to normalization trace
- 20 mW in, no atten, 1308 nm, Set to normalization trace
- 25 mW in, no atten, 1308 nm, Set to normalization trace
- 30 mW in, no atten, 1308 nm, Set to normalization trace
- 40 mW in, no atten, 1308 nm, Set to normalization trace
- 50 mW in, no atten, 1308 nm, Set to normalization trace
- 60 mW in, no atten, 1308 nm, Set to normalization trace
- 40 mW in, no atten, 1309.8 nm, Set to normalization trace
- 70 mW in, no atten, 1309.8 nm, Set to normalization trace

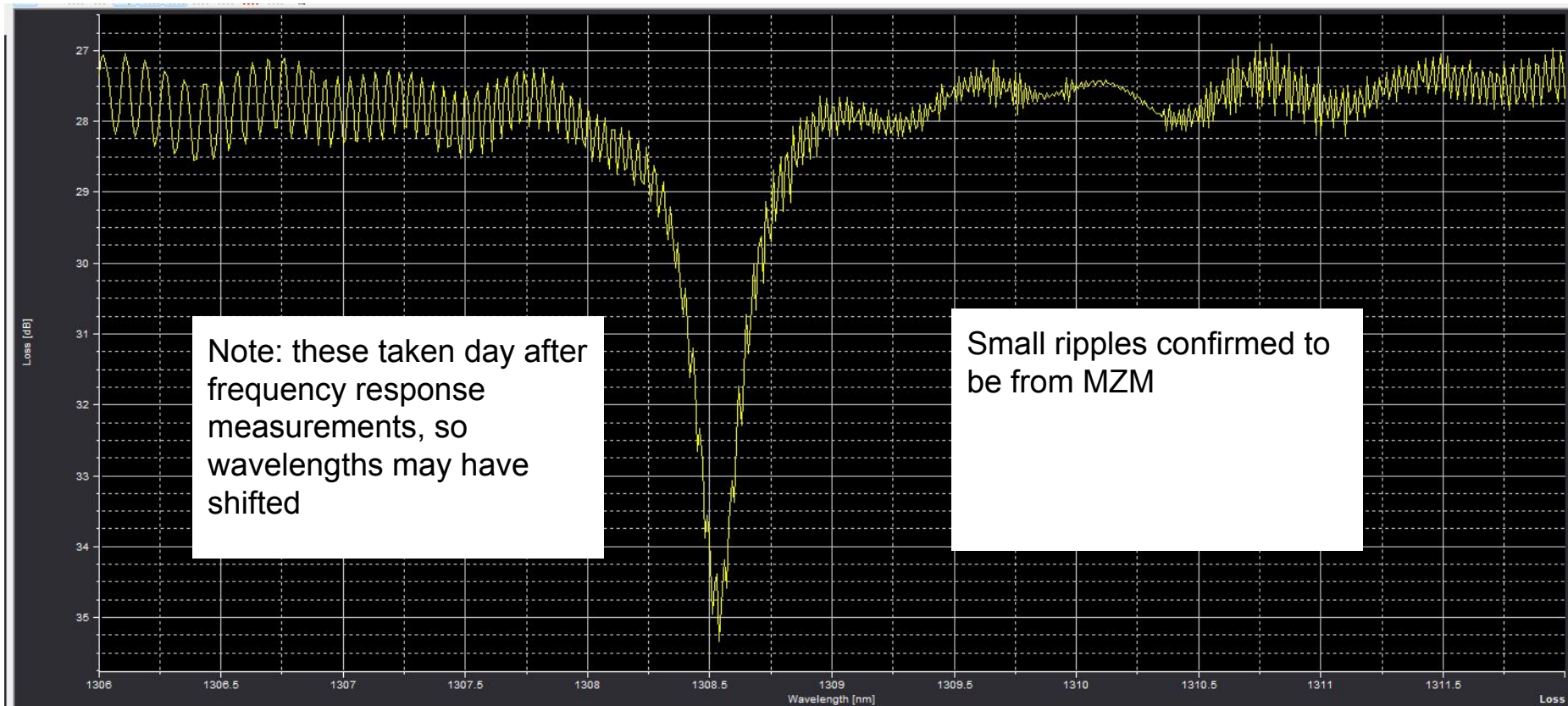
Results – Wavelength sweeps

HERE WILL GO CALIBRATED WAVELENGTH SWEEPS VS. POWER

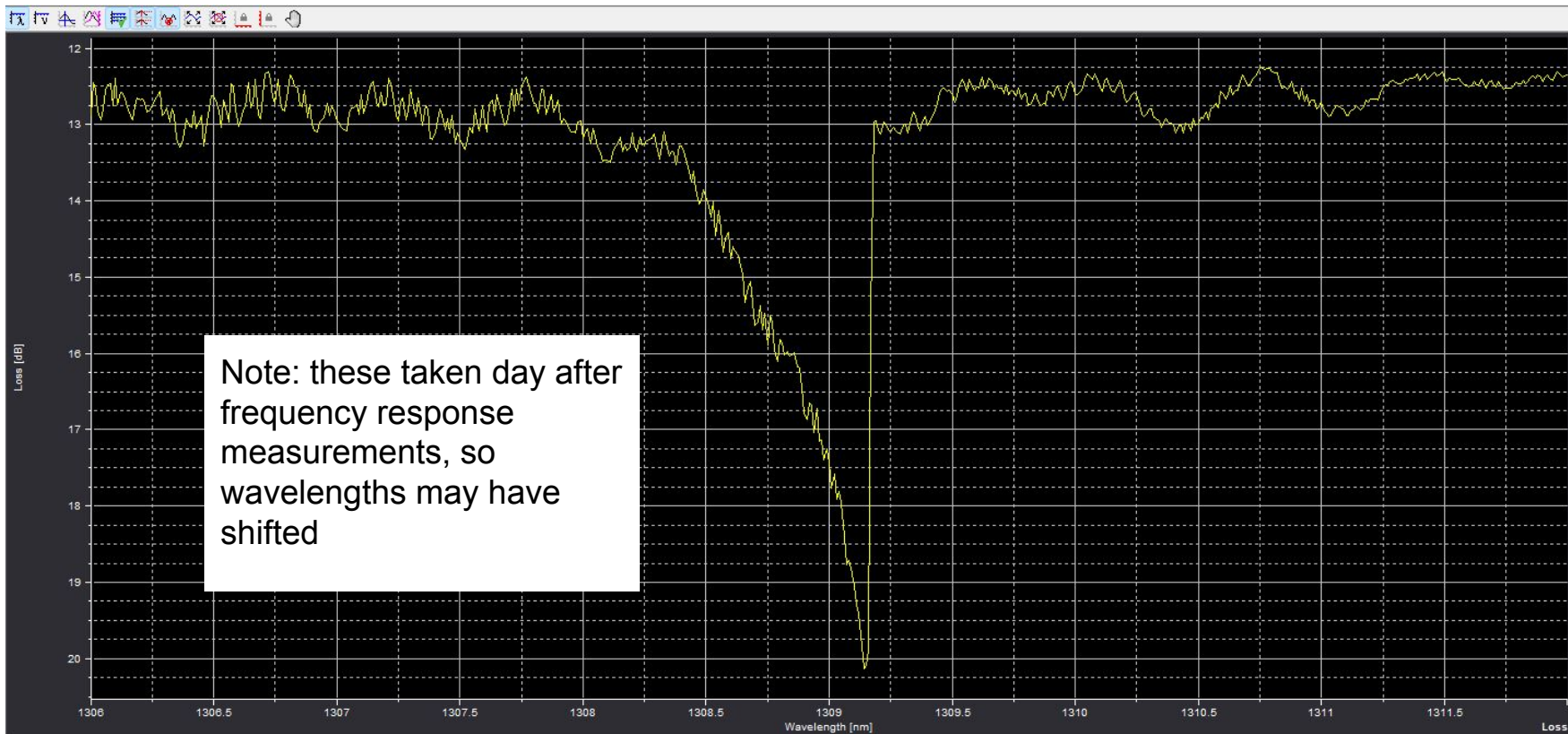
Note for the time being: confirmed on scope that we could see DC value change, consistent with going through MRM resonance in “no” self-heating regime and with self-heating present

TODO: Get estimate for gain of self-heating loop

Results – Wavelength sweeps

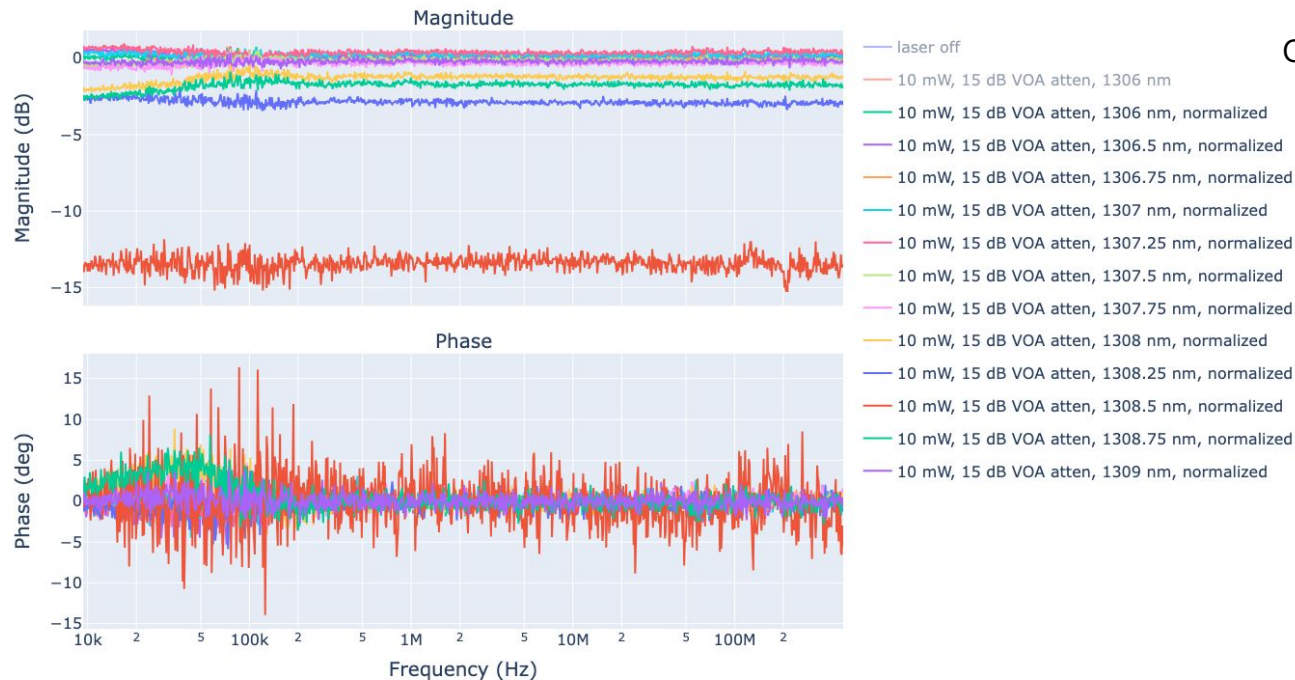


Results – Wavelength sweeps



Results – Low-Power Cavity Response

Bode sweep overlay

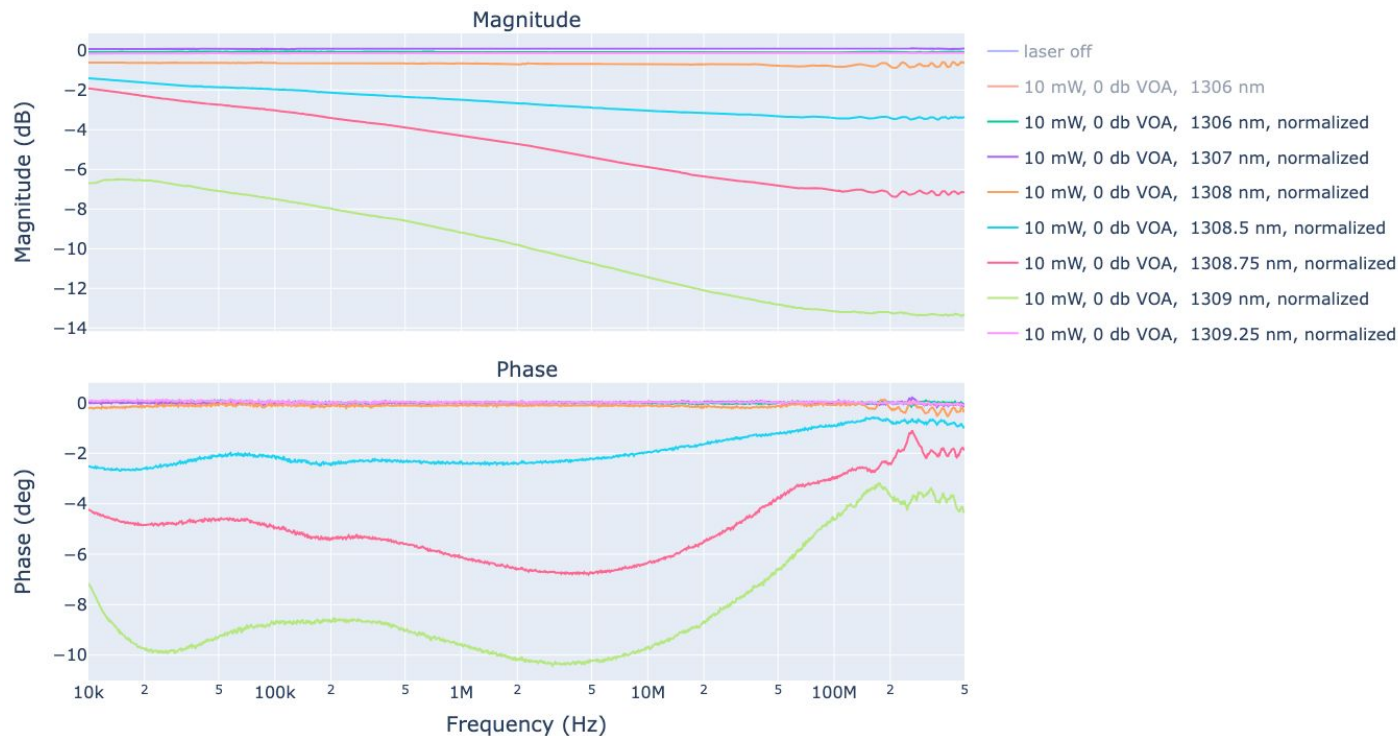


Conclusions:

- Confirmed modulating far below cavity linewidth, as cavity does not impart phase on reflection
- Response dips as through port DC power dips, as expected

Results – High-Power Cavity Response

Bode sweep overlay



Normalized to MRM bank response when off-resonance

Conclusions:

- Some high-power effect with frequency dependence
- Not quite a full pole/zero
- **Any ideas what is happening?**

TODO: normalize these by photodetector DC value

Discussion

Next Steps

Scale plots appropriately

Carefully control detuning, try to get to maximum self-heating gain term

Try with biased PN Junction in MRM

Measure how effects scale with power

The two points above should allow us to distinguish between carrier / NLA effects vs. pure thermal

Focus more on high-frequency response

Normalize by DC value of PD